

Report Gasoline

Time Series - MESIO UPC-UB

Abdelrahman Eissa & Benjamín Adasme

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Introduction

- Gasoline consumption in Spain and the world: factors
- Dataset Gasoline
- Objectives

Methodology

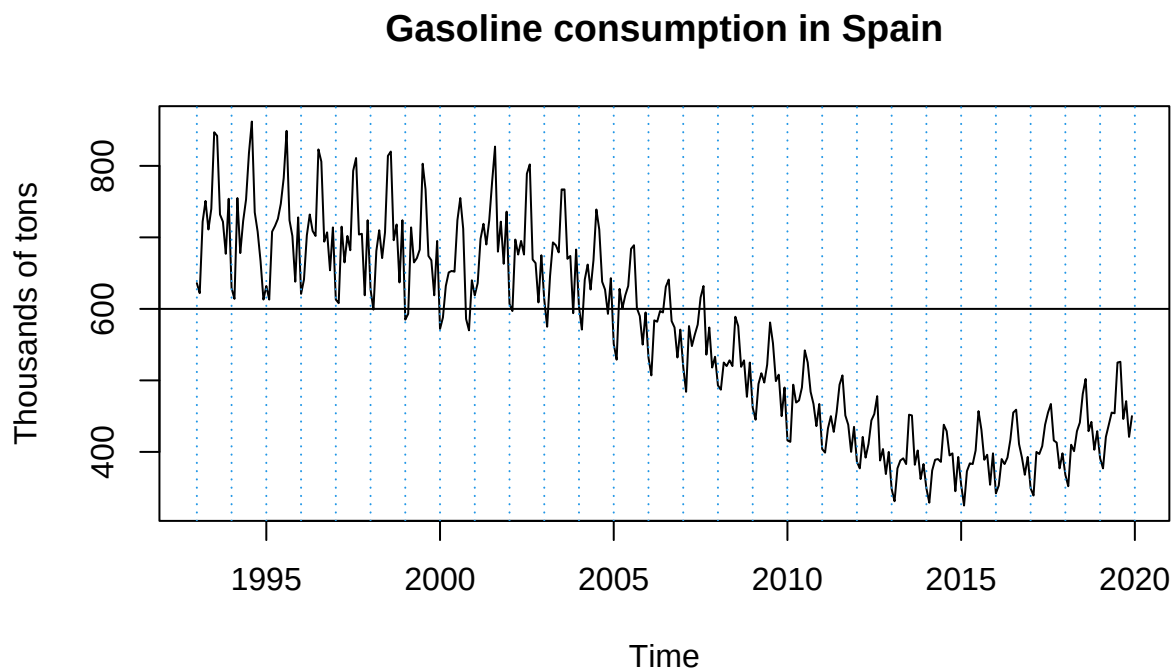
Explain the methodology (steps, approaches, etc.)

- Classic ARIMA: Box-jenkins methodology for time series
- ARIMAX: calendar effects and outlier treatment.

Results and discussion

Classic ARIMA approach

Model identification



The model has a mean of 563.19 and a variance of 136.34. Since we detect this time serie is not stationary, we applied the next transformations:

- Box-Cox transformation:
- Seasonality:
- Non-constant variance:

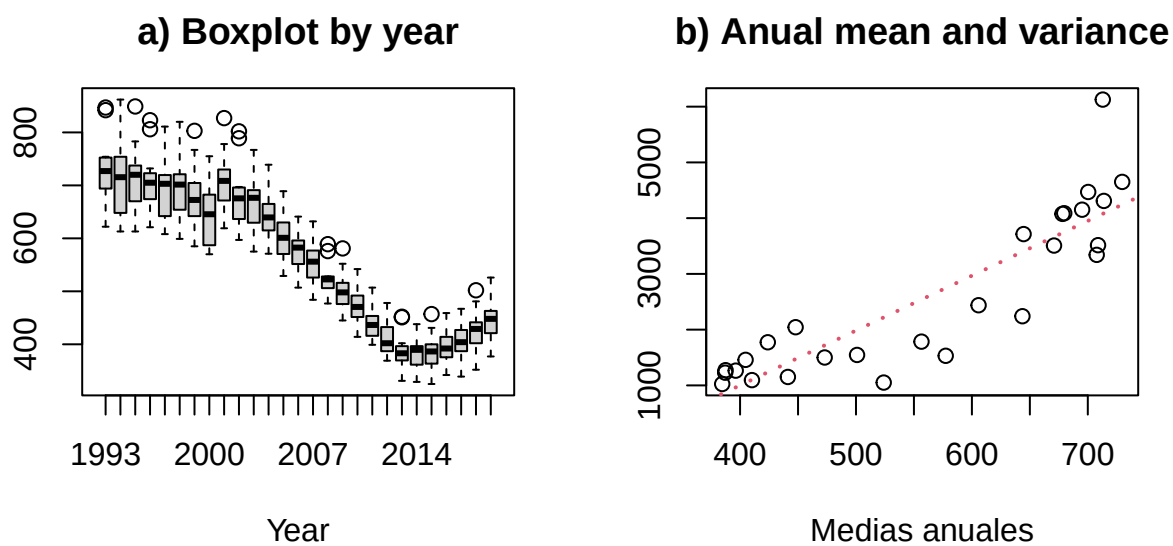
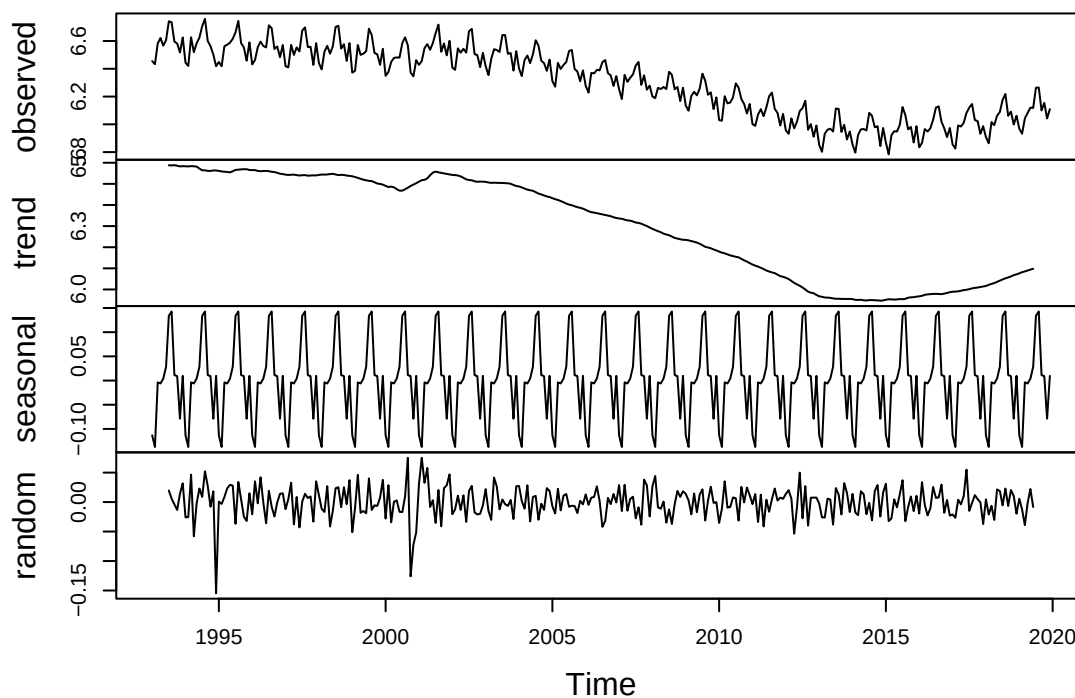
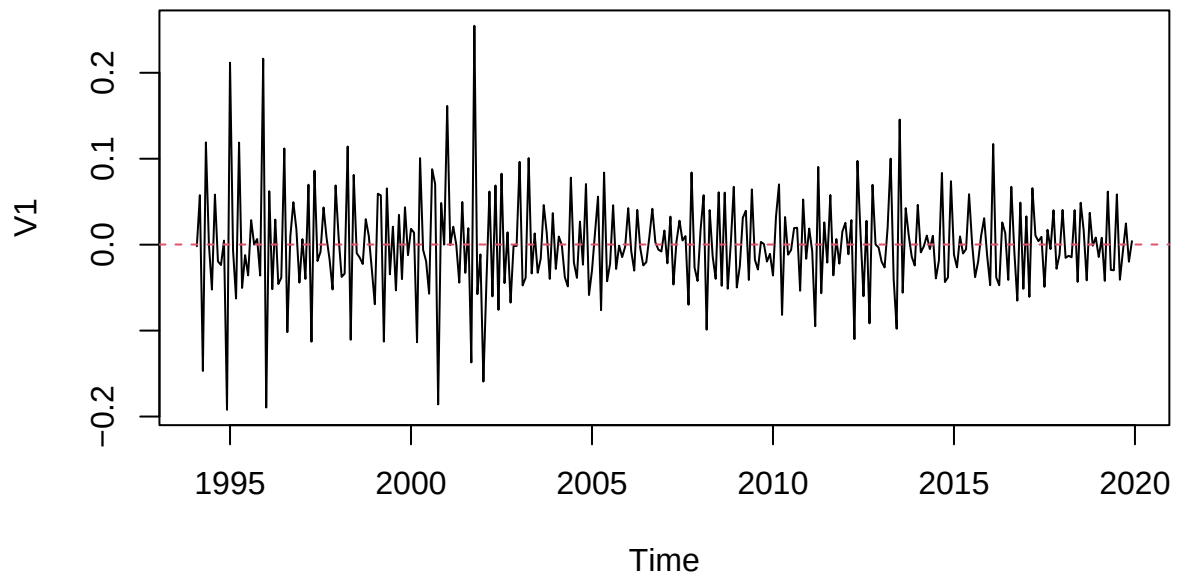


Figure 1: Non-constant variance serie

Decomposition of additive time series

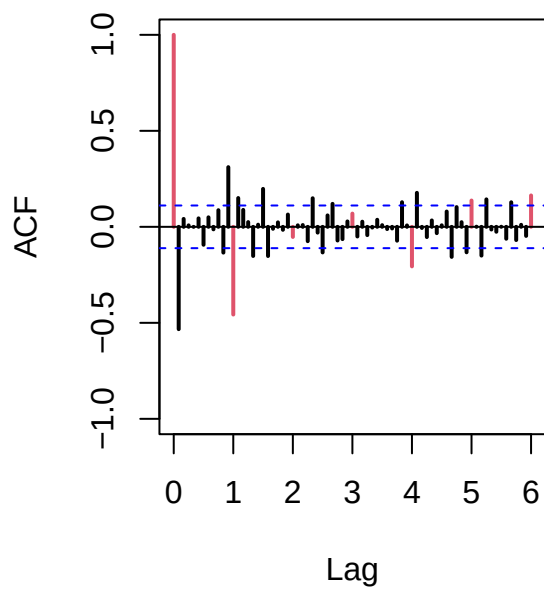


Stationary series – Mean: 0

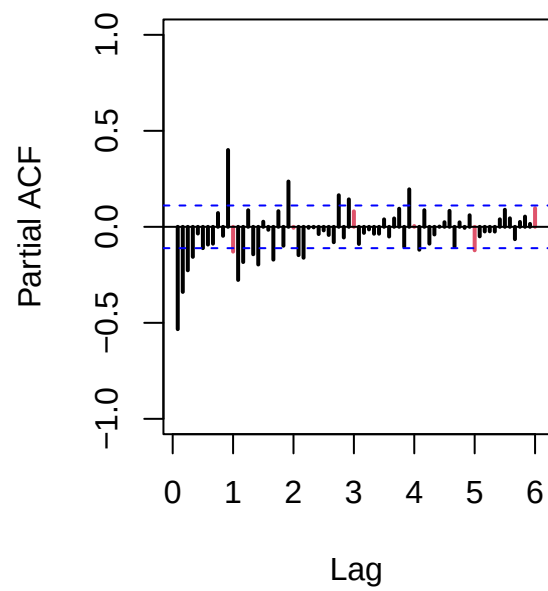


Estimation for plausible models

ACF



PACF



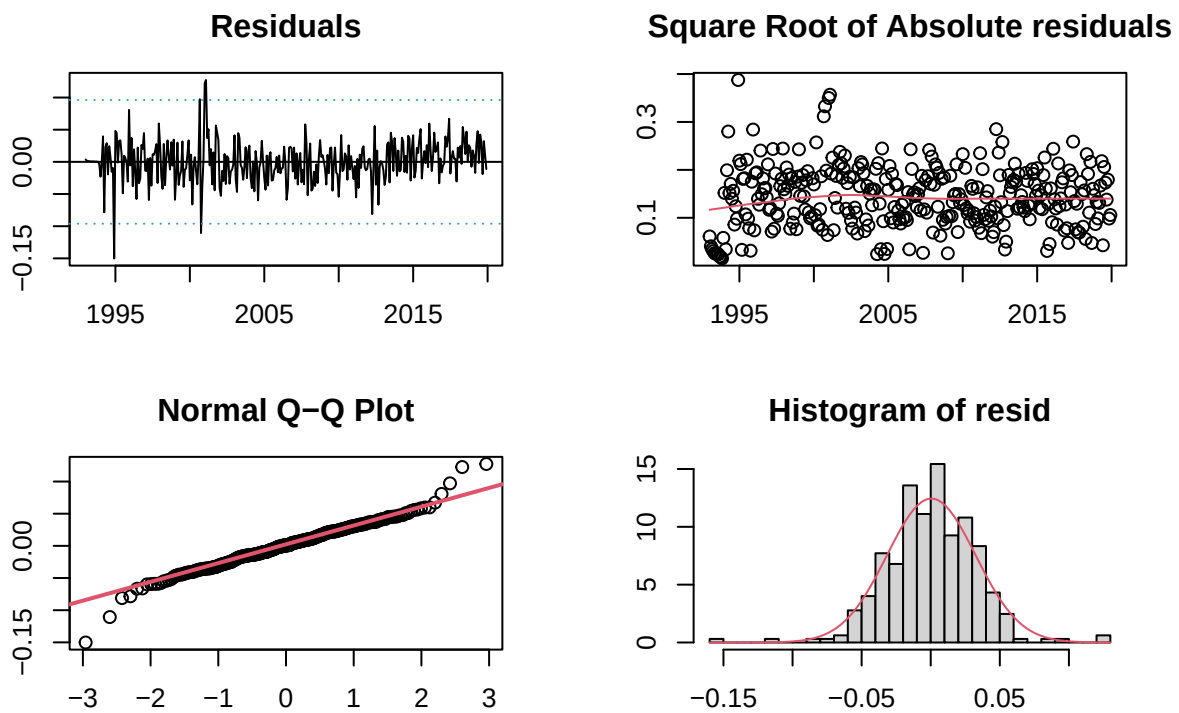
Seasonal part: MA (1)

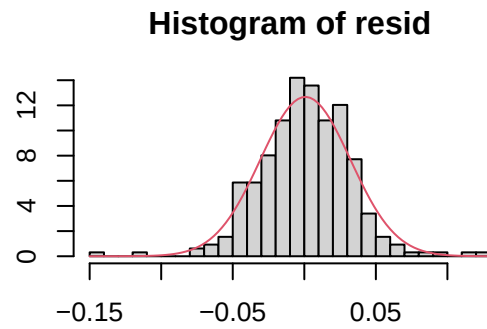
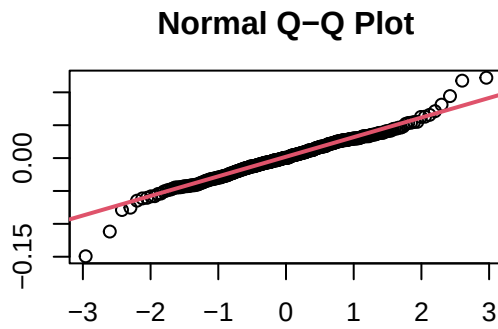
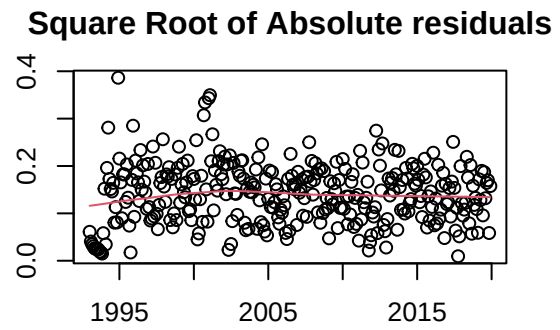
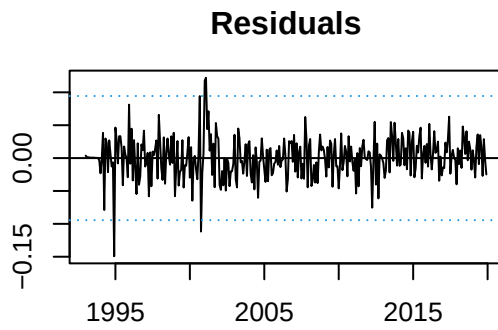
Regular part:

- MA(1)
- ARMA(1,1)
- MA(2)
- ARMA(1,2)

Models	Log-Likelihood	AIC	AR(1)	MA(1)	MA(2)	SMA(1)
MA(1)	610.97	-1215.93	NA	-0.68	NA	-0.89
ARMA(1,1)	613.46	-1218.93	-0.17	-0.60	NA	-0.88
MA(2)	613.95	-1219.89	NA	-0.79	0.14	-0.88
ARMA(1,2)	617.75	-1225.50	0.99	-1.73	0.75	-0.92

Model validation





Prediction assesment and forecasting

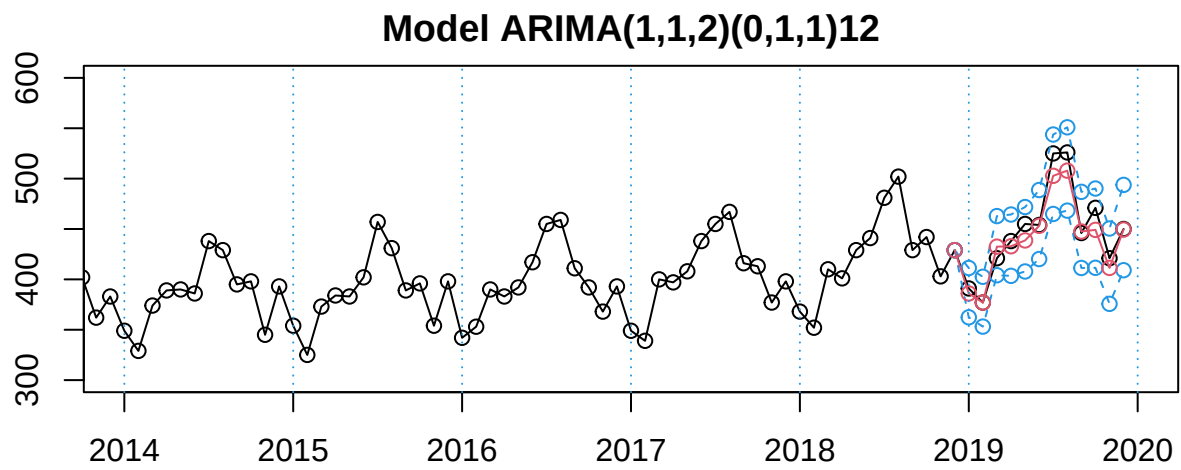
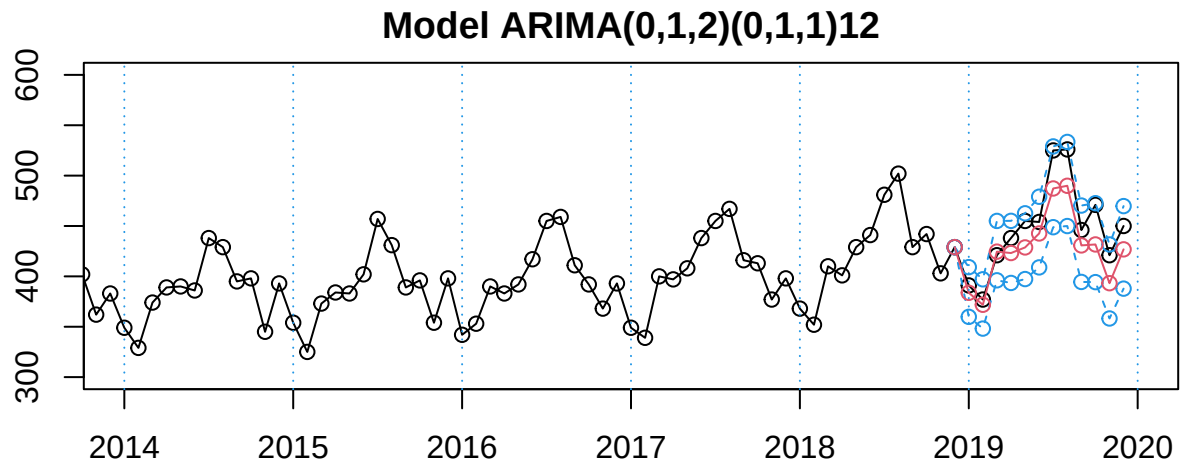
- Both models are stable

MA(2) stability

Variables	Full	Cut	Diference
ma1	-0.79	-0.78	0.01
ma2	0.14	0.13	0.01
sma1	-0.88	-0.89	0.01

ARMA(1,2) stability

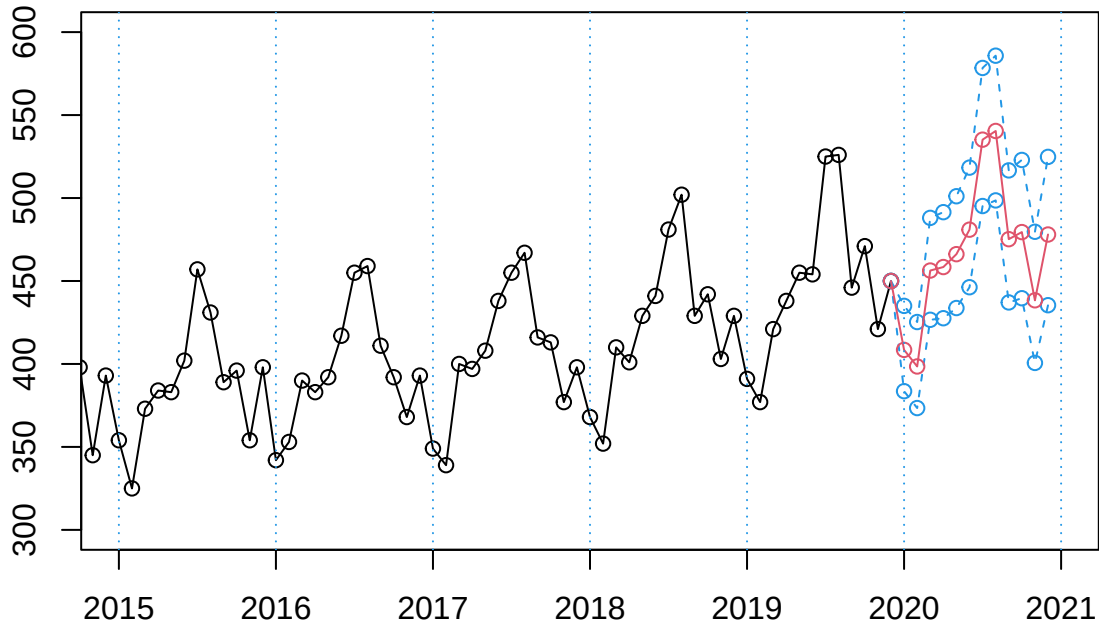
Variables	Full	Cut	Diference
ar1	0.99	0.99	0.00
ma1	-1.73	-1.72	0.01
ma2	0.75	0.74	0.01
sma1	-0.92	-0.93	0.01



Model	RMSE	MAE	RMSPE	MAPE	Mean CI
MA(2)	24.06	20.66	0.05	0.04	63.65
ARMA(1,2)	12.40	9.39	0.03	0.02	63.52

Selected model: **ARMA(1,2)**

Model ARIMA(1,1,2)(0,1,1)12



ARIMAX Analysis: Outlier and Calendar Effects Treatment

```
##
## Call:
## arima(x = lnserie, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12),
##       xreg = data.frame(wTradDays, wEast))
##
## Coefficients:
##          ar1          ma1          ma2          sma1  wTradDays    wEast
##          0.9890    -1.6805    0.6980    -0.9019         4e-03    0.0304
## s.e.    0.0149    0.0424    0.0408    0.0419         5e-04    0.0077
##
## sigma^2 estimated as 0.0008347:  log likelihood = 651.32,  aic = -1288.64
##
##          ar1          ma1          ma2          sma1  wTradDays    wEast
## 66.242567 -39.617071  17.128261 -21.522962   8.108000   3.943221
```

term	estimate	std.error
ar1	0.9889695	0.0149295
ma1	-1.6805023	0.0424186
ma2	0.6980143	0.0407522
sma1	-0.9018532	0.0419019
wTradDays	0.0040107	0.0004947
wEast	0.0304024	0.0077100

```
## [1] 0.0003537106
```


	Obs	type_detected	W_coeff	ABS_L_Ratio	Fecha	perc.Obs
10	16	AO	-0.055	2.968	Abr 1994	-0.055
2	24	AO	-0.152	6.876	Dic 1994	-0.152
7	36	AO	0.064	3.313	Dic 1995	0.064
9	38	TC	0.054	3.171	Feb 1996	0.054
19	79	AO	0.045	2.674	Jul 1999	0.045
3	93	LS	0.092	5.596	Sep 2000	0.092
1	94	TC	-0.143	6.650	Oct 2000	-0.143
4	97	LS	0.069	4.325	Ene 2001	0.069
15	98	AO	0.046	2.641	Feb 2001	0.046
8	108	AO	0.064	3.366	Dic 2001	0.064
14	164	AO	-0.047	2.662	Ago 2006	-0.047
11	171	AO	0.049	2.694	Mar 2007	0.049
18	182	AO	0.046	2.691	Feb 2008	0.046
12	186	TC	-0.045	2.723	Jun 2008	-0.045
16	219	LS	-0.036	2.681	Mar 2011	-0.036
5	232	LS	-0.054	3.428	Abr 2012	-0.054
6	234	AO	0.081	4.106	Jun 2012	0.081
17	278	AO	0.046	2.631	Feb 2016	0.046
13	294	AO	0.048	2.632	Jun 2017	0.048

ARIMA and ARIMAX models comparison

Conclusions

Appendices

Validation function

```
#####Validation#####
validation=function(model,dades){
  s=frequency(get(model$series))
  resid=model$residuals
  par(mfrow=c(2,2),mar=c(3,3,3,3))
  #Residuals plot
  plot(resid,main="Residuals")
  abline(h=0)
  abline(h=c(-3*sd(resid),3*sd(resid)),lty=3,col=4)
  #Square Root of absolute values of residuals (Homocedasticity)
  scatter.smooth(sqrt(abs(resid)),main="Square Root of Absolute residuals",
    lpars=list(col=2))

  #Normal plot of residuals
  qqnorm(resid)
  qqline(resid,col=2,lwd=2)

  ##Histogram of residuals with normal curve
  hist(resid,breaks=20,freq=FALSE)
  curve(dnorm(x,mean=mean(resid),sd=sd(resid)),col=2,add=T)

  #ACF & PACF of residuals
  par(mfrow=c(1,2))
  acf(resid,ylim=c(-1,1),lag.max=60,col=c(2,rep(1,s-1)),lwd=1)
  pacf(resid,ylim=c(-1,1),lag.max=60,col=c(rep(1,s-1),2),lwd=1)
  par(mfrow=c(1,1))

  #ACF & PACF of square residuals
  par(mfrow=c(1,2))
  acf(resid^2,ylim=c(-1,1),lag.max=60,col=c(2,rep(1,s-1)),lwd=1)
  pacf(resid^2,ylim=c(-1,1),lag.max=60,col=c(rep(1,s-1),2),lwd=1)
  par(mfrow=c(1,1))

  #Ljung-Box p-values
  par(mar=c(2,2,1,1))
  tsdiag(model,gof.lag=7*s)
  cat("\n-----\n")
  print(model)

  #Stationary and Invertible
  cat("\nModul of AR Characteristic polynomial Roots: ",
    Mod(polyroot(c(1,-model$model$phi))),"\n")
  cat("\nModul of MA Characteristic polynomial Roots: ",
    Mod(polyroot(c(1,model$model$theta))),"\n")

  #Model expressed as an MA infinity (psi-weights)
  psis=ARMAtoMA(ar=model$model$phi,ma=model$model$theta,lag.max=36)
  names(psis)=paste("psi",1:36)
  cat("\nPsi-weights (MA(inf))\n")
```

```

cat("\n-----\n")
print(psis[1:20])

#Model expressed as an AR infinity (pi-weights)
pis=-ARMAtoMA(ar=-model$model$theta,ma=-model$model$phi,lag.max=36)
names(pis)=paste("pi",1:36)
cat("\nPi-weights (AR(inf))\n")
cat("\n-----\n")
print(pis[1:20])

## Add here complementary tests (use with caution!)
##-----
cat("\nNormality Tests\n")
cat("\n-----\n")

##Shapiro-Wilks Normality test
print(shapiro.test(resid(model)))

suppressMessages(require(nortest,quietly=TRUE,warn.conflicts=FALSE))
##Anderson-Darling test
print(ad.test(resid(model)))

suppressMessages(require(tseries,quietly=TRUE,warn.conflicts=FALSE))
##Jarque-Bera test
print(jarque.bera.test(resid(model)))

cat("\nHomoscedasticity Test\n")
cat("\n-----\n")
suppressMessages(require(lmtest,quietly=TRUE,warn.conflicts=FALSE))
##Breusch-Pagan test
obs=get(model$series)
print(bptest(resid(model)~I(obs-resid(model))))

cat("\nIndependence Tests\n")
cat("\n-----\n")

##Durbin-Watson test
print(dwtest(resid(model)~I(1:length(resid(model)))))

##Ljung-Box test
cat("\nLjung-Box test\n")
print(t(apply(matrix(c(1:4,(1:4)*s)),1,function(e1) {
  te=Box.test(resid(model),type="Ljung-Box",lag=e1)
  c(lag=(te$parameter),statistic=te$statistic[[1]],p.value=te$p.value)})))

#Sample ACF vs. Teoric ACF
par(mfrow=c(2,2),mar=c(3,3,3,3))
acf(dades, ylim=c(-1,1),lag.max=36,main="Sample ACF")

plot(ARMAacf(model$model$phi,model$model$theta,lag.max=36),ylim=c(-1,1),
      type="h",xlab="Lag", ylab="", main="ACF Teoric")
abline(h=0)

```

```

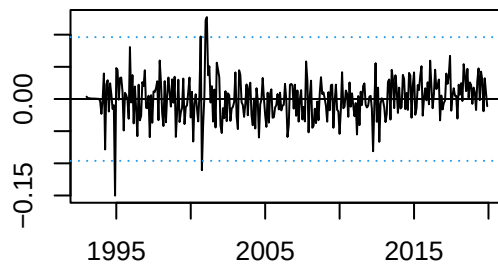
#Sample PACF vs. Teoric PACF
pacf(dades, ylim=c(-1,1) ,lag.max=36,main="Sample PACF")

plot(ARMAacf(model$model$phi,model$model$theta,lag.max=36, pacf=T),ylim=c(-1,1),
      type="h", xlab="Lag", ylab="", main="PACF Teoric")
abline(h=0)
par(mfrow=c(1,1))
}
##### Fi Validation #####

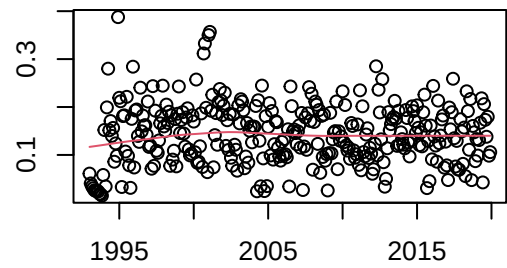
```

ARIMA models validation detailed

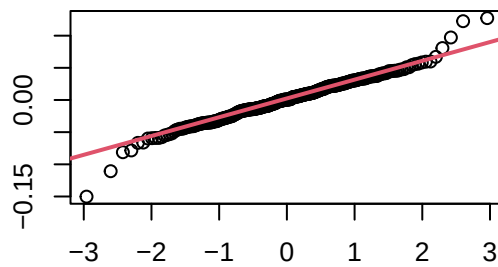
Residuals



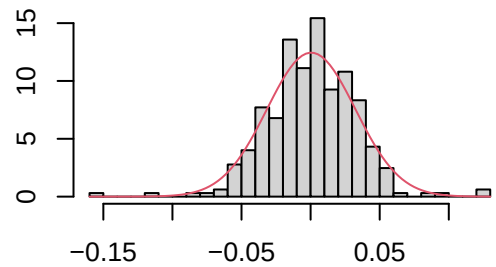
Square Root of Absolute residuals



Normal Q-Q Plot

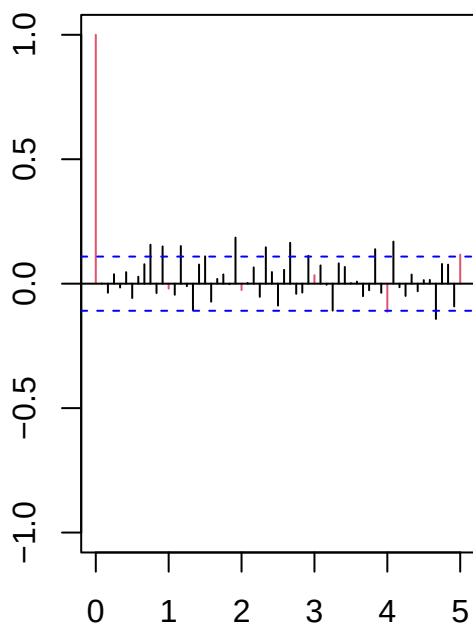


Histogram of resid

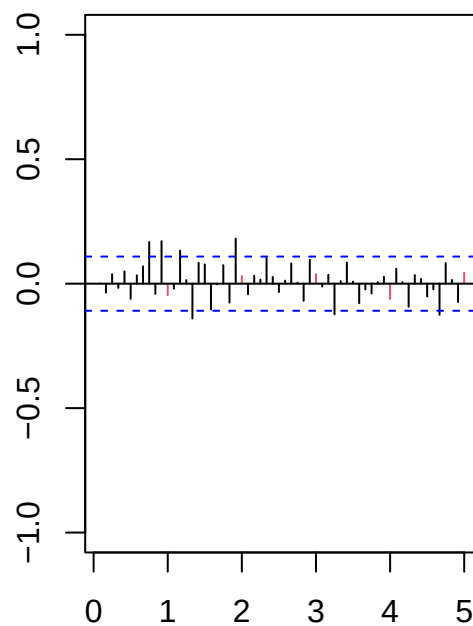


MA(2)

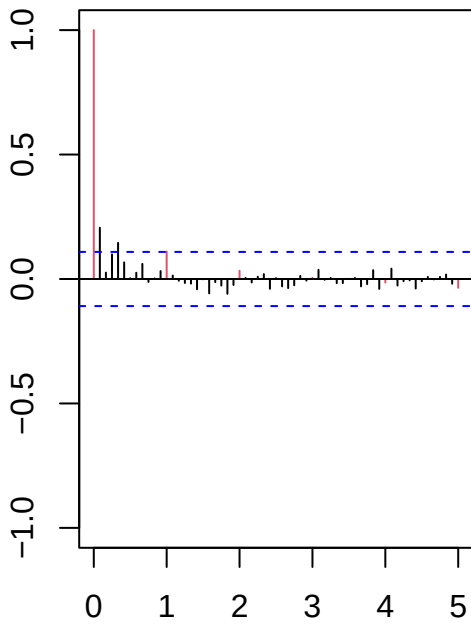
Series resid



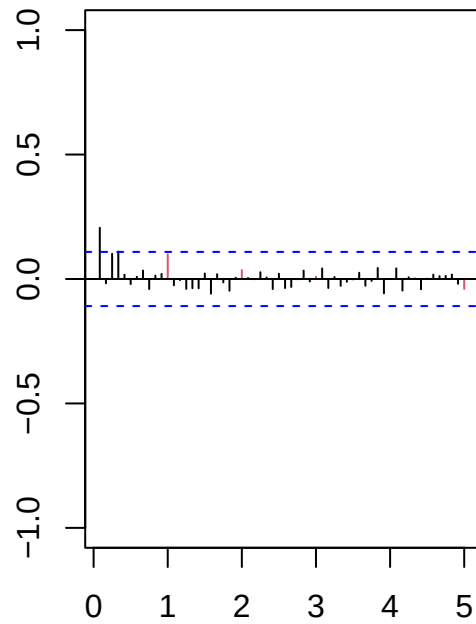
Series resid



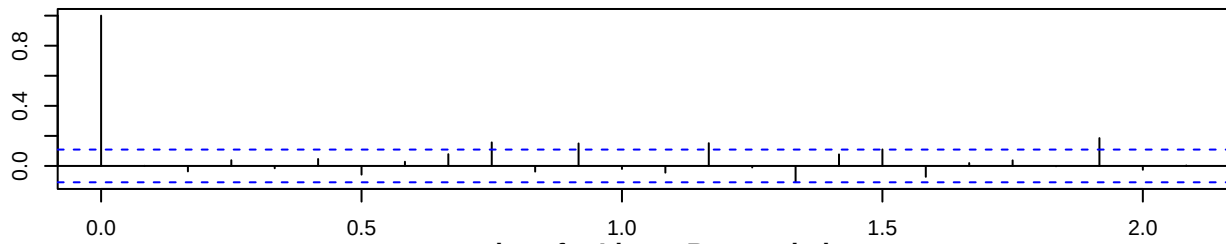
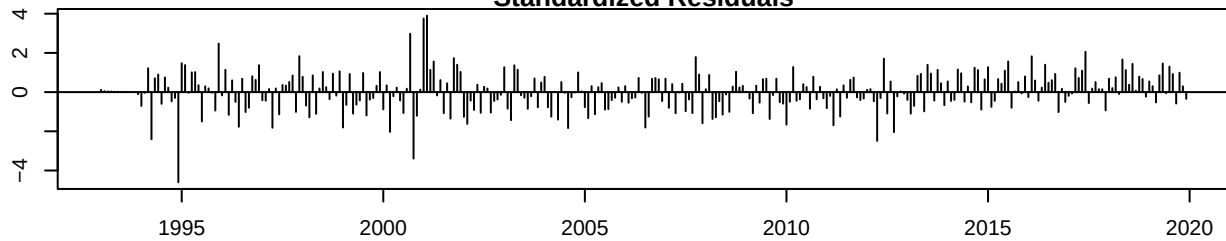
Series resid^2



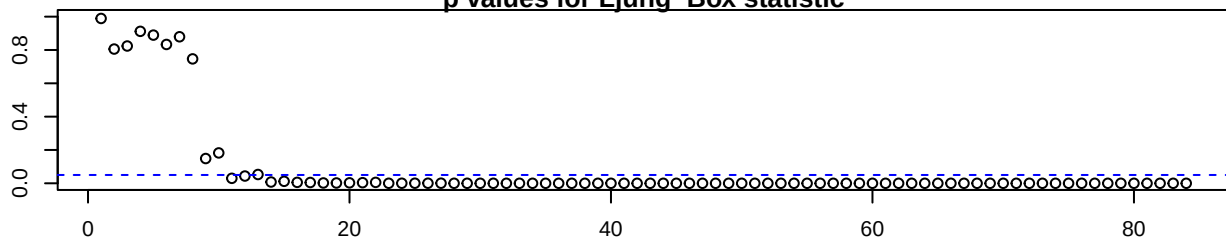
Series resid^2



Standardized Residuals



p values for Ljung-Box statistic



```
##
## -----
##
## Call:
## arima(x = lnserie, order = c(0, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
```

```

##
## Coefficients:
##      ma1      ma2      sma1
##    -0.7872  0.1387 -0.8759
## s.e.   0.0595  0.0556  0.0451
##
## sigma^2 estimated as 0.001065:  log likelihood = 613.95,  aic = -1219.89
##
## Modul of AR Characteristic polynomial Roots:
##
## Modul of MA Characteristic polynomial Roots:  1.0111 1.0111 1.0111 1.0111 1.0111 1.0111 1.0111 1.0111
##
## Psi-weights (MA(inf))
##
## -----
##      psi 1      psi 2      psi 3      psi 4      psi 5      psi 6      psi 7
## -0.7872180  0.1386801  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000
##      psi 8      psi 9      psi 10     psi 11     psi 12     psi 13     psi 14
##  0.0000000  0.0000000  0.0000000  0.0000000 -0.8759337  0.6895508 -0.1214746
##      psi 15     psi 16     psi 17     psi 18     psi 19     psi 20
##  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000
##
## Pi-weights (AR(inf))
##
## -----
##      pi 1      pi 2      pi 3      pi 4      pi 5      pi 6
## -0.787218034 -0.481032141 -0.269505707 -0.145450171 -0.077125922 -0.040543873
##      pi 7      pi 8      pi 9      pi 10     pi 11     pi 12
## -0.021221038 -0.011082956 -0.005781767 -0.003014526 -0.001571273 -0.876752591
##      pi 13     pi 14     pi 15     pi 16     pi 17     pi 18
## -0.689977547 -0.421574638 -0.236185008 -0.127465088 -0.067588657 -0.035530140
##      pi 19     pi 20
## -0.018596766 -0.009712386
##
## Normality Tests
##
## -----
##
## Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.97599, p-value = 3.076e-05
##
## Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.73796, p-value = 0.05399
##
## Jarque Bera Test
##
## data:  resid(model)
## X-squared = 71.851, df = 2, p-value = 2.22e-16

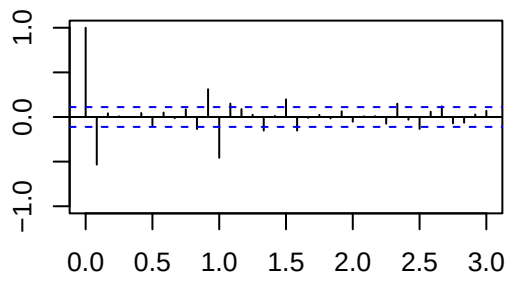
```

```

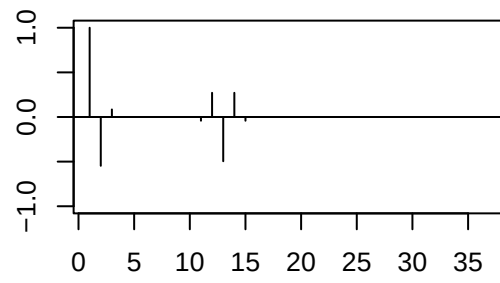
##
##
## Homoscedasticity Test
##
## -----
##
## studentized Breusch-Pagan test
##
## data: resid(model) ~ I(obs - resid(model))
## BP = 2.3425, df = 1, p-value = 0.1259
##
##
## Independence Tests
##
## -----
##
## Durbin-Watson test
##
## data: resid(model) ~ I(1:length(resid(model)))
## DW = 2.0239, p-value = 0.5635
## alternative hypothesis: true autocorrelation is greater than 0
##
##
## Ljung-Box test
##      lag.df      statistic      p.value
## [1,]      1 1.830573e-04 9.892051e-01
## [2,]      2 4.321965e-01 8.056561e-01
## [3,]      3 9.062973e-01 8.239079e-01
## [4,]      4 9.863685e-01 9.118559e-01
## [5,]     12 2.149488e+01 4.358670e-02
## [6,]     24 5.472970e+01 3.403871e-04
## [7,]     36 8.541745e+01 6.762500e-06
## [8,]     48 1.098332e+02 9.496701e-07

```

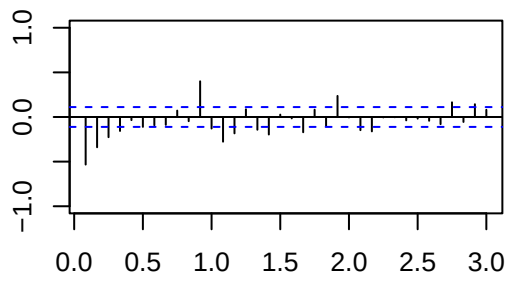

Sample ACF



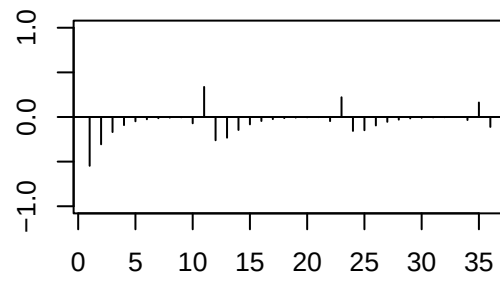
ACF Teoric

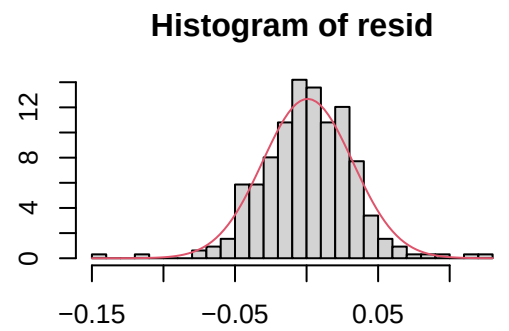
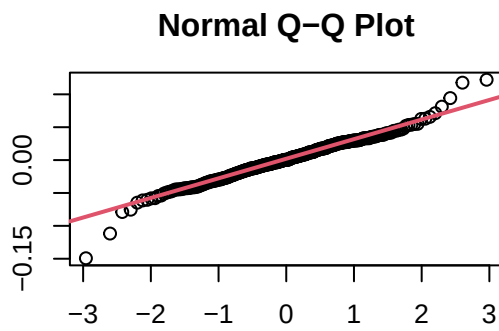
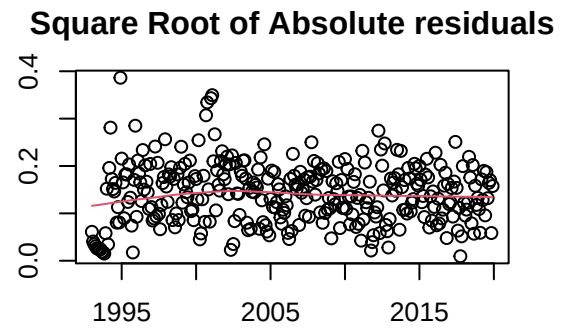
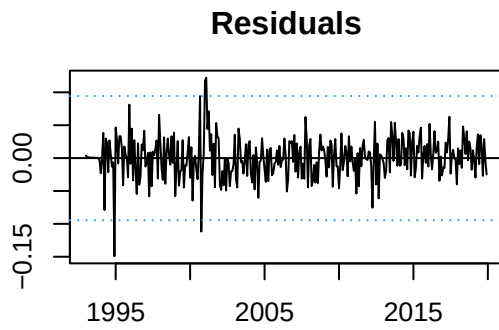


Sample PACF



PACF Teoric

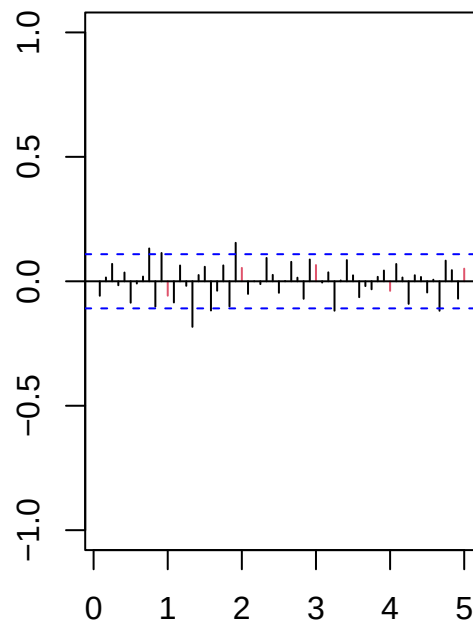
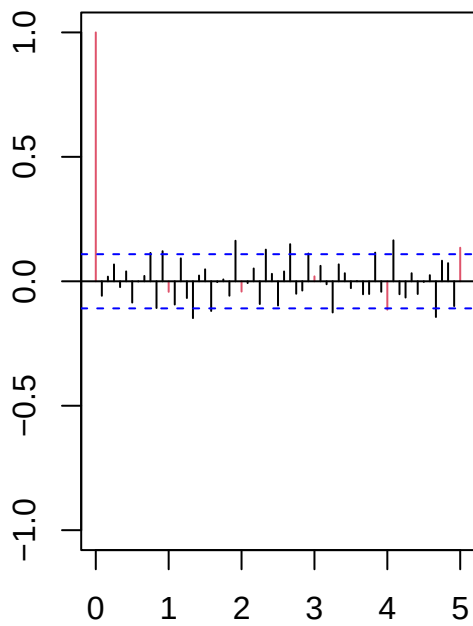




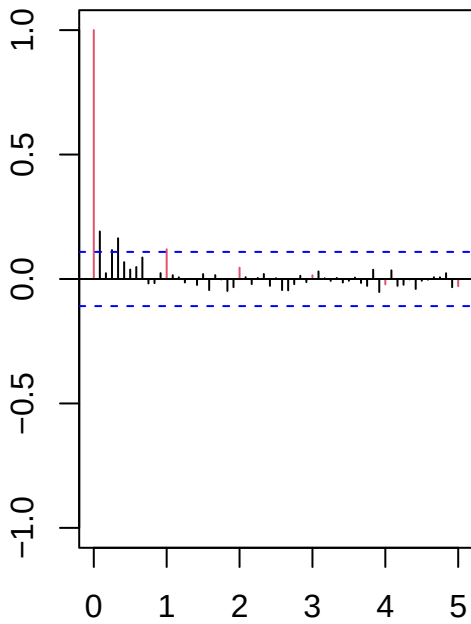
ARMA(1,2)

Series resid

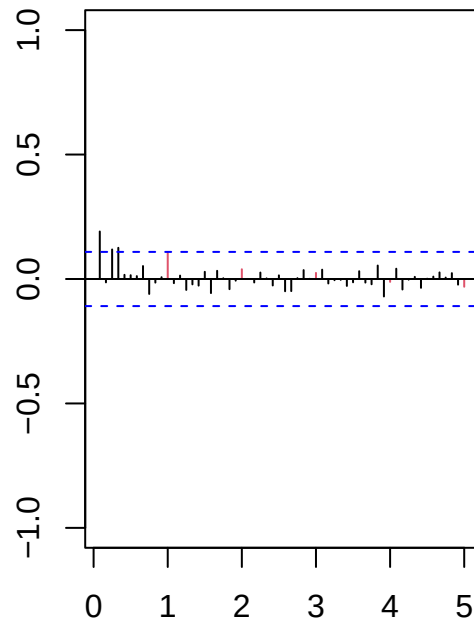
Series resid



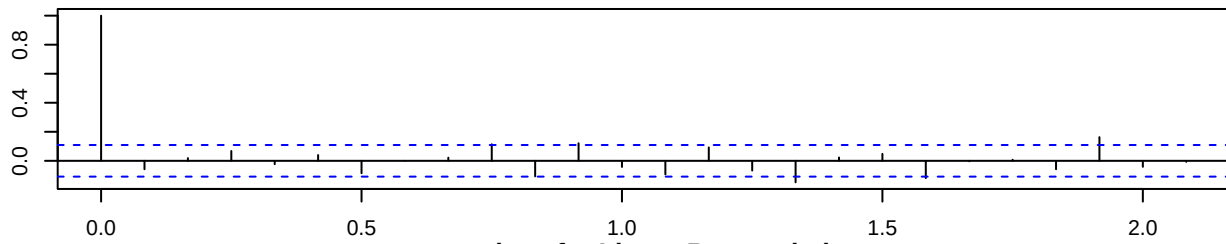
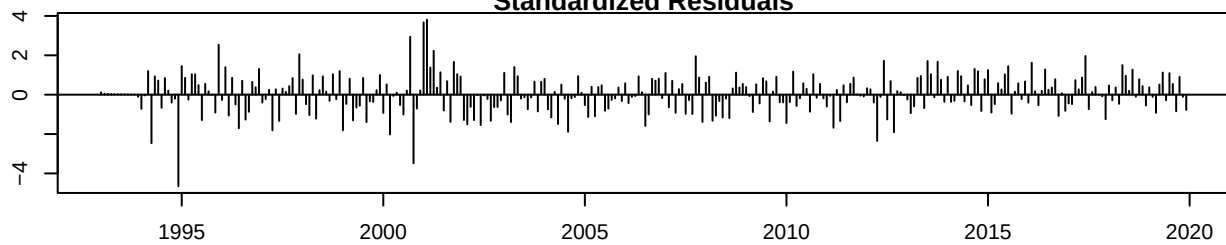
Series resid^2



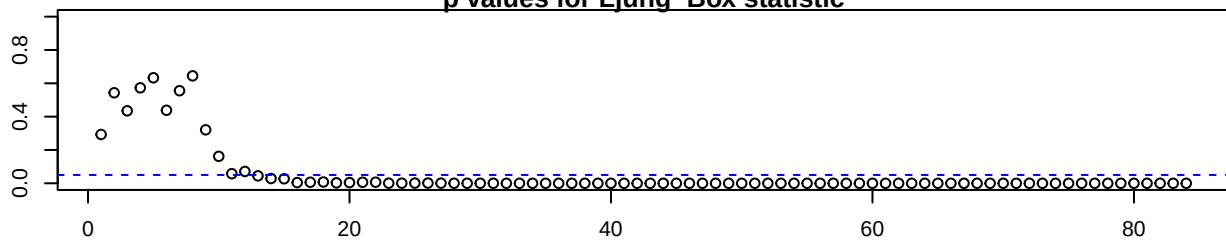
Series resid^2



Standardized Residuals



p values for Ljung-Box statistic



```
##
## -----
##
## Call:
## arima(x = lnserie, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
```

```

##
## Coefficients:
##      ar1      ma1      ma2      sma1
##      0.9888 -1.7281  0.7451 -0.9219
## s.e.  0.0159  0.0377  0.0356  0.0452
##
## sigma^2 estimated as 0.001028:  log likelihood = 617.75,  aic = -1225.5
##
## Modul of AR Characteristic polynomial Roots:  1.011333
##
## Modul of MA Characteristic polynomial Roots:  1.006796 1.006796 1.006796 1.006796 1.006796 1.006796
##
## Psi-weights (MA(inf))
##
## -----
##      psi 1      psi 2      psi 3      psi 4      psi 5
## -0.7393437608  0.0140457561  0.013883529  0.0137327135  0.0135788183
##      psi 6      psi 7      psi 8      psi 9      psi 10
##  0.0134266478  0.0132761825  0.0131274034  0.0129802916  0.0128348284
##      psi 11     psi 12     psi 13     psi 14     psi 15
##  0.0126909953 -0.9093917674  0.6940391340 -0.0006802568 -0.0006726335
##      psi 16     psi 17     psi 18     psi 19     psi 20
## -0.0006650957 -0.0006576423 -0.0006502724 -0.0006429852 -0.0006357796
##
## Pi-weights (AR(inf))
##
## -----
##      pi 1      pi 2      pi 3      pi 4      pi 5      pi 6
## -0.739343761 -0.532583440 -0.369489249 -0.241698053 -0.142379471 -0.065961066
##      pi 7      pi 8      pi 9      pi 10     pi 11     pi 12
## -0.007902253  0.035491682  0.067222500  0.089724712  0.104968862 -0.807394186
##      pi 13     pi 14     pi 15     pi 16     pi 17     pi 18
## -0.561891887 -0.369433617 -0.219764071 -0.104515990 -0.016870873  0.048720107
##      pi 19     pi 20
##  0.096765590  0.130922674
##
## Normality Tests
##
## -----
##
## Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.9754, p-value = 2.419e-05
##
## Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.81635, p-value = 0.03455
##
## Jarque Bera Test
##

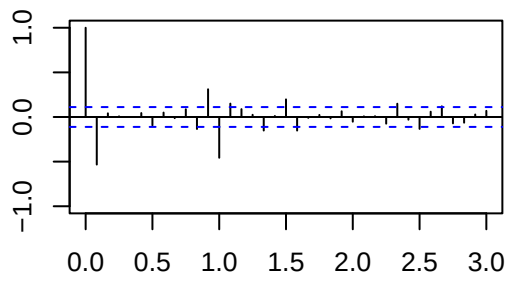
```

```

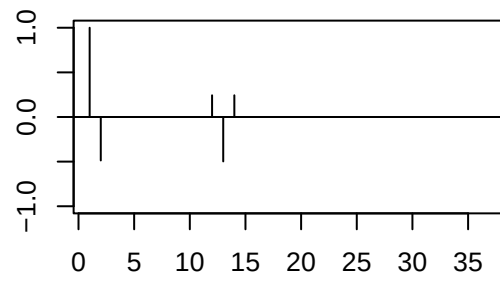
## data: resid(model)
## X-squared = 73.466, df = 2, p-value < 2.2e-16
##
##
## Homoscedasticity Test
##
## -----
##
## studentized Breusch-Pagan test
##
## data: resid(model) ~ I(obs - resid(model))
## BP = 3.1928, df = 1, p-value = 0.07397
##
##
## Independence Tests
##
## -----
##
## Durbin-Watson test
##
## data: resid(model) ~ I(1:length(resid(model)))
## DW = 2.1273, p-value = 0.8629
## alternative hypothesis: true autocorrelation is greater than 0
##
##
## Ljung-Box test
##      lag.df  statistic      p.value
## [1,]      1   1.105301 2.931057e-01
## [2,]      2   1.220740 5.431499e-01
## [3,]      3   2.730671 4.350402e-01
## [4,]      4   2.910458 5.729201e-01
## [5,]     12  19.796407 7.103661e-02
## [6,]     24  51.747428 8.439303e-04
## [7,]     36  79.893649 3.581262e-05
## [8,]     48 102.282063 8.457064e-06

```

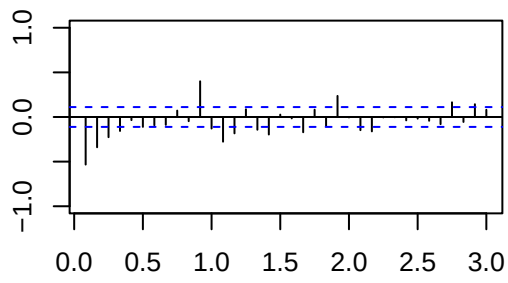
Sample ACF



ACF Teoric



Sample PACF



PACF Teoric

